

Advanced (Carolina Reaper)

- Q1.** What is the purpose of graphing a system of linear equations?
 (A) To find the slope of each equation
 (B) To identify the intersection point as the solution
 (C) To determine the y -intercept of each equation
 (D) To find the equation of the lines in slope-intercept form
 (E) To identify the type of system (independent or dependent)
- Q2.** What does the point of intersection represent in a graph of a system of linear equations?
 (A) The x -intercept of the first equation
 (B) The y -intercept of the second equation
 (C) The solution to the system of equations
 (D) The slope of the first equation
 (E) The slope of the second equation
- Q3.** Which of the following systems of linear equations has no solution?
 (A) $x + y = 2$ and $x + y = 3$
 (B) $x - y = 1$ and $x + y = 2$
 (C) $2x + 2y = 4$ and $x + y = 2$
 (D) $x + 2y = 3$ and $2x + 4y = 6$
 (E) $3x - 2y = 1$ and $x + 2y = 3$
- Q4.** Which of the following is a correct statement about the graph of the system $x + y = 2$ and $x - y = 1$?
 (A) The lines intersect at $(1, 1)$
 (B) The lines intersect at $(2, 0)$
 (C) The lines are parallel
 (D) The lines coincide
 (E) The lines have the same slope
- Q5.** What is the x -coordinate of the point of intersection of the lines $y = 2x - 1$ and $y = -x + 2$?
 (A) $\frac{1}{3}$
 (B) $\frac{2}{3}$
 (C) $\frac{3}{2}$
 (D) $\frac{4}{3}$
- Q6.** Which of the following systems of linear inequalities has a solution set that is a triangle?
 (A) $x + y \geq 2$ and $x - y \leq 1$ and $y \geq 0$
 (B) $x + y \leq 2$ and $x - y \leq 1$ and $y \geq 0$
 (C) $x + y \geq 2$ and $x - y \geq 1$ and $y \leq 0$
 (D) $x + y \leq 2$ and $x - y \geq 1$ and $y \leq 0$
 (E) $x + y \geq 2$ and $x - y \leq 1$ and $y \leq 0$
- Q7.** Which of the following is a correct statement about the graph of the system $x + y = 2$ and $y = -x + 2$?
 (A) The lines intersect at $(0, 2)$
 (B) The lines intersect at $(1, 1)$
 (C) The lines are parallel
 (D) The lines coincide
 (E) The lines have the same slope
- Q8.** What does it mean for two lines to be parallel in the context of a system of linear equations?
 (A) The lines intersect at a single point
 (B) The lines do not intersect
 (C) The lines have the same slope
 (D) The lines have the same y -intercept
 (E) The lines are perpendicular

Open Ended Questions (FRQ)

- Q9.** Graph the system of linear equations $x + y = 2$ and $x - y = 1$. Identify the point of intersection and interpret the solution.
- Q10.** Graph the system of linear inequalities $x + y \geq 2$ and $x - y \leq 1$ and $y \geq 0$. Describe the solution set.

Chef's Tips

- Tip 1: Make sure students understand the concept of intersection points as solutions to systems of linear equations.
- Tip 2: Emphasize the importance of accuracy when graphing systems of linear equations.
- Tip 3: Encourage students to interpret graphical solutions in the context of real-world problems.